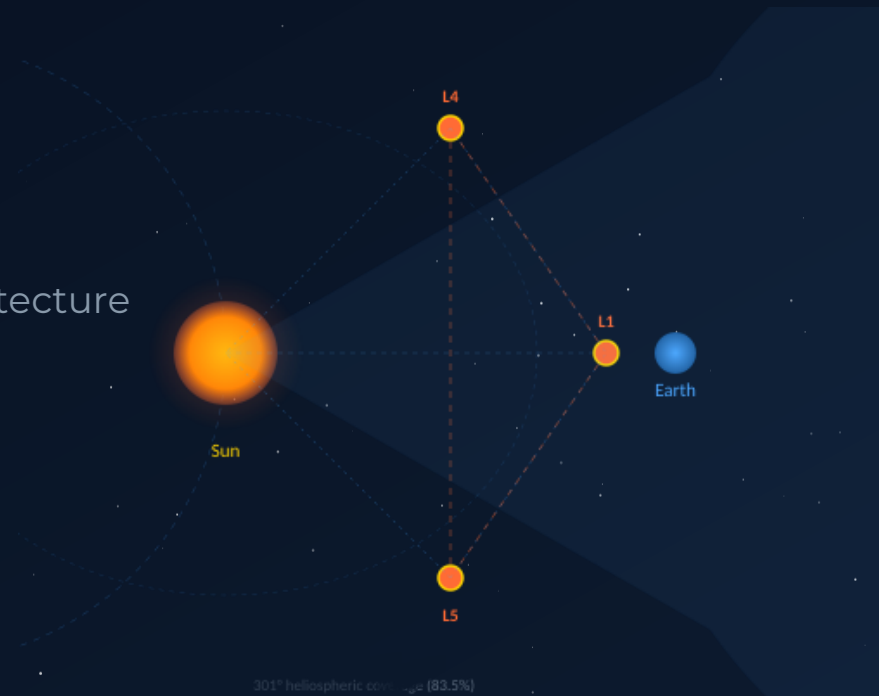


HELIOS

Autonomous Solar Radiation Warning Architecture
for Human Deep-Space Operations

Paweł Micał | Space AI 2026



"Protecting Human Life Beyond Earth's Magnetosphere"

! THE PROBLEM

"August 4, 1972 — between Apollo 16 and 17 — a solar storm delivered a lethal dose to deep space. Had astronauts been on the Moon, they would have received 4,000 mSv. The NASA limit is 250 mSv."

15-60 min Warning

Current systems detect CME magnetic structure only when it physically passes L1. Insufficient for crew protective action.

Earth-Centric Design

All existing systems require Earth-based manual analysis. No autonomous onboard crew warning capability exists.

Limited Coverage

Single-point L1 observation. Less than 50% solar surface monitored. Major geometric blind spots.

985 mSv

Bastille Day dose
(394% NASA limit)

40 sats

Starlink loss
2022 storm

58,000

Satellites by 2030
(10× current)

\$42B

Potential daily losses
from severe storm

THE SOLUTION — HELIOS



Distributed Architecture

L1 (validation) + L4/L5 (stereoscopic observation). Three-node constellation spanning 301° of the heliosphere.



Autonomous AI Onboard

Neural network infers CME magnetic structure (Bz). No Earth dependency. TMR fault-tolerant voting.



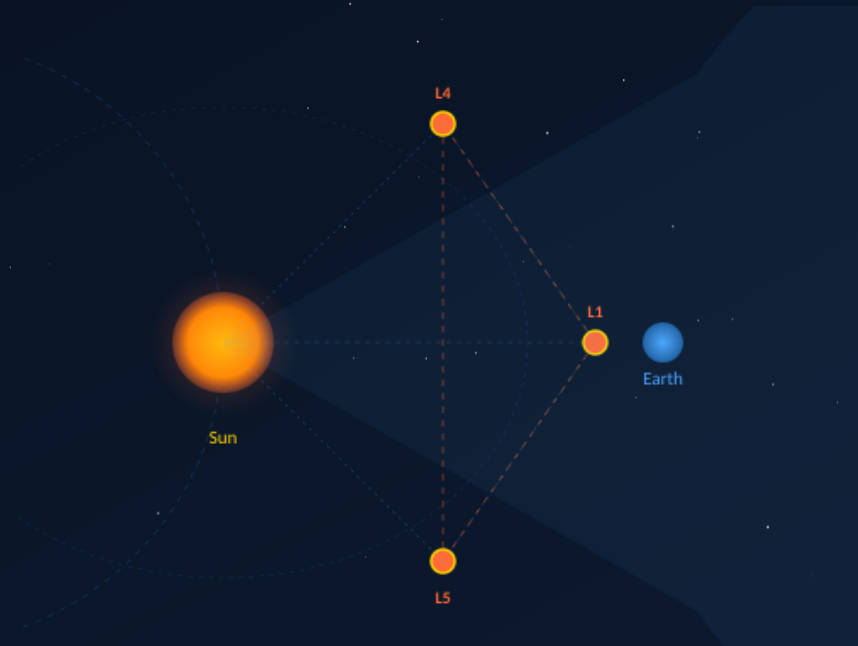
16-50 Hour Warning

Hours of advance notice vs. 15-60 minutes today. Speed-dependent: 50h for 800 km/s, 16h for 2500 km/s CME.



Crew-Centric Design

Dose-based warnings in NASA-standard units (mSv). Severity classification: Low / Moderate / High / Extreme.



83.3%

Heliospheric coverage
3.5-10.9h earlier detection
vs single-point L1

HOW IT WORKS

01

DETECT

0.2–7.2h

Multi-perspective coronagraph imaging across L1/L4/L5. Running-difference algorithm triggers pipeline.

83.3% severity accuracy

02

TRIANGULATE

1–16h

Stereoscopic 3D reconstruction via Monte Carlo triangulation. Overdetermined 3-satellite geometry.

1.1M km resolution

03

INFER Bz

Onboard

Ensemble neural network (3 seeds) predicts magnetic field orientation. Heteroscedastic uncertainty.

6.3 nT MAE

04

WARN CREW

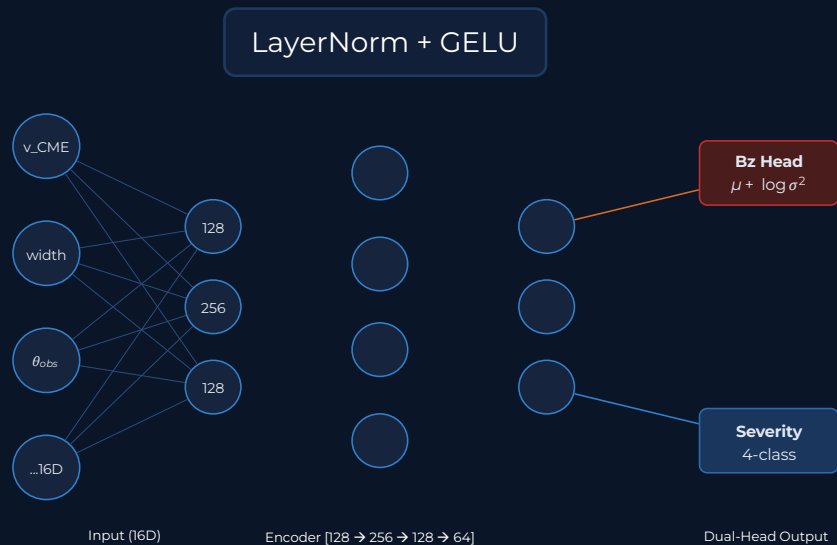
1–10 min

TMR voting consensus. Physics-based dosimetry calculation. Autonomous severity classification.

83.3% accuracy

First system to predict CME magnetic structure BEFORE it reaches L1

AI/ML ARCHITECTURE



Feature Vector

16-dimensional: v_CME, width, θ_{obs} , parallax, triangulation quality, ...

Encoder

[16→128→256→128→64] with LayerNorm + GELU activation

Bz Head

Heteroscedastic regression: outputs $\mu(Bz) + \log \sigma^2$ for uncertainty

Severity Head

4-class classifier: Low / Moderate / High / Extreme

Loss Function

Multi-task: $\alpha=0.5$ (Gaussian NLL) + $\beta=0.5$ (Cross-Entropy)

Ensemble

3-seed training (42, 123, 456). Mean aggregation. 2.7 nT spread

Training Pipeline

10,000 synthetic events (physics-based flux rope models) + 19 historical CMEs (12 train, 6 test, 1 unseen showcase) with 50× oversampling = 10,600 training samples

Innovation: Physics-informed architecture · Uncertainty quantification · Fault-tolerant consensus · No data leakage

✓ VALIDATED PERFORMANCE

Bastille Day 2000 — Showcase Event

Predicted Bz: **-54.9 nT**

Actual Bz: **-60.0 nT**

Error: 5.2 nT (8.6%)

Predicted Dose: 985 mSv (394% NASA 250 mSv SPE limit)

Classification: EXTREME ✓ (correct)

Aggregate Validation Metrics

Bastille Day Confidence **91.6%**

Improvement vs Baseline **36%**

Bz MAE (aggregate) **6.3 nT**

Bz MAE (Bastille Day) **5.2 nT**

Severity Accuracy **83.3%**

Adjacent-or-Correct **100%**

Validation Methodology

- ▶ 6 independent historical events (test set) — zero data leakage between train/test splits
- ▶ 3-seed ensemble with mean aggregation — inter-model spread: 2.7 nT (robust consensus)
- ▶ Physics-based dosimetry formula calibrated on 8 historical SPEs (1972–2017)
- ▶ Full pipeline: open-source GitHub repository — every metric independently reproducible

ORBITAL MECHANICS

L1 Halo Orbit Parameters

Ax amplitude	220,908 km
Ay amplitude	705,322 km
Az amplitude	249,993 km
Period	177.74 days
SEZ min. separation	9.51° (compliant)
Stability index (ν)	804 (active control req.)
HOI ΔV	~50 m/s
Station-keeping	4.0 m/s/year
12-year propellant	17.2 kg

Insertion Innovation

Large-amplitude L1 Halo Orbit enables stable manifold insertion — dramatically reducing HOI delta-V vs direct insertion. L4/L5 nodes use Tadpole Orbit Insertion (TOI).

HELIOS L1 ~50 m/s

HELIOS L4/L5 ~50 m/s

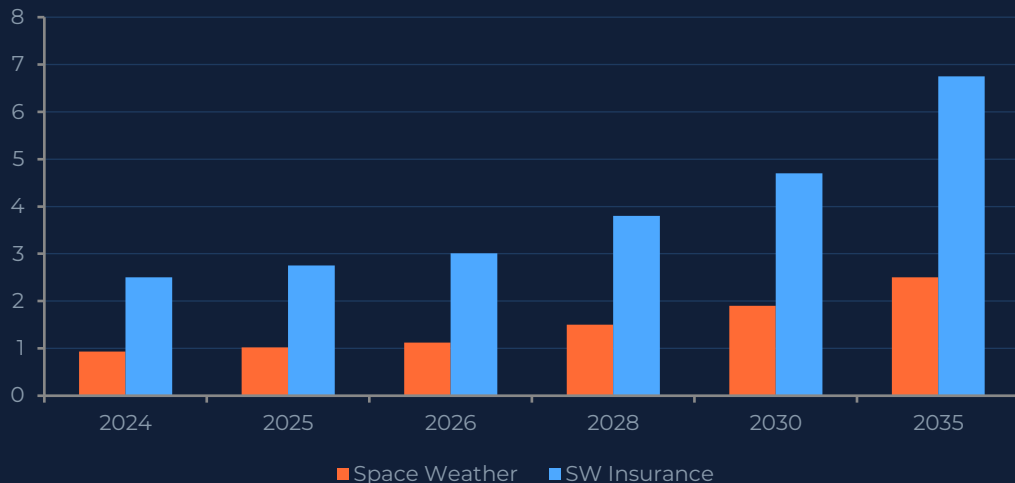
Constellation Delta-V Budget

Node	HOI/TOI	SK/yr	Propellant
L1	~50 m/s	4.0 m/s	17.2 kg
L4	~50 m/s	1.5 m/s	14.3 kg
L5	~50 m/s	1.5 m/s	14.3 kg

Validated: Differential correction • Monodromy matrix stability

MARKET OPPORTUNITY

Market Growth (\$B)



WHY NOW?

- ▶ **Artemis II** — 2026: First crewed lunar flight since Apollo
- ▶ **SOLAR-1** operational mid-2026 — still single-point, still 15-60 min warning
- ▶ **58,000** satellites projected by 2030
- ▶ **Munich Re** (Jan 2026): New space weather insurance products launched
- ▶ **JHU-APL**: "2026 will see greater focus on space weather prediction"
- ▶ Focus on **space datacenters** (Starcloud, SpaceX)

\$2.5B

Space Weather
Market 2035

\$6.75B

SW Insurance
Market 2035

\$374B

Space Infrastructure
Market 2034

58,000

Satellites
by 2030



TARGET CUSTOMER SEGMENTS

TIER 1 — Crew Safety (Life-Critical)

Moon Bases

SpaceX HLS, NASA Artemis, ESA

\$100K–\$1M/mission

Private Stations

Axiom Space, Vast, Blue Origin

\$500K–\$5M/yr

Space Tourism

SpaceX Crew Dragon, Axiom Space, Vast

\$10K–\$100K/flight

TIER 2 — Asset Protection

Mega-Constellations

Starlink, Kuiper, OneWeb

\$500K–\$5M/yr

Space Datacenters

SpaceX, Starcloud

\$1M+/yr

Space Power Grids

Starcatcher, SSP operators

\$1M+/yr

TIER 3 — Terrestrial & Science

Aviation

Airlines, Boeing, Airbus

\$50K–\$500K/yr

Scientific Data

NASA, ESA, JAXA, universities

\$100K–\$1M/yr

Insurance

Munich Re, Tokio Marine, Swiss Re

Revenue share

Value Chain Logic:

Artemis crew needs → Protection mandate → Station operators adopt → Insurance requires → Constellation operators follow → Market standard

COMPETITIVE LANDSCAPE

Capability	HELIOS	Mission Space	ESA Vigil	NOAA SOLAR-1
Architecture	L1+L4+L5 distributed	LEO constellation	Single L5	Single L1
Bz Prediction	✓ < 6 nT MAE	✗ Not possible	✗ No (In-situ at L5)	✗ No (In-situ only)
Onboard AI	✓ TMR voting	✗ None	✗ None	✗ None
Crew Warnings	✓ Autonomous	✗ None	✗ None	✗ None
Warning Time	16-50 hours	"96h" (unvalidated)	4-5 days (rotation)	15-60 minutes
Peer-Reviewed	Planned (Q2 2026)	✗ No validation	✓ ESAPhase (C/D)	✓ NASA
Hardware in Space	✗ None	✓ ZOHAR-I (LEO)	✗ None (2031 launch)	✓ Operational

HELIOS Unique Positioning

- ▶ Only system designed for autonomous deep-space crew protection (vs Earth-centric approaches)
- ▶ Distributed L1+L4+L5 observation — new data from L4/L5 currently unavailable globally
- ▶ Early magnetic structure (Bz) inference — hours before L1 in-situ measurement (physically impossible for competitors)

\$ BUSINESS MODEL & REVENUE

Government Contracts

35%

NASA/ESA partnerships, DoD/Space Force

\$2-10M/yr

Data Subscription

40%

Tiered: Basic alerts → Full API → Enterprise integration

\$5-20M/yr

Mission Integration

15%

Spacecraft/habitat integration, per-mission licensing

\$2-10M/yr

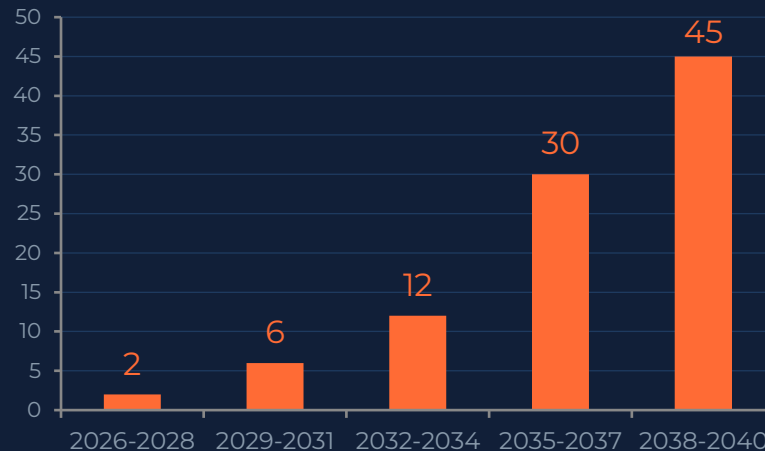
Insurance Intel

10%

Risk data for Munich Re, Swiss Re, Tokio Marine HCC

\$1-5M/yr

Revenue Projection (\$M)



Cost Structure (Estimates)

Satellite constellation (3 nodes) **\$50-150M** (phased over 3 years)

Launch (rideshare per node) **\$5-15M** (SpaceX Falcon 9)

Ground operations/year **\$2-5M** (3-5 global stations)

Team (20-40 people)/year **\$3-7M** (engineering + BD)



GO-TO-MARKET STRATEGY

PHASE 1

2026–2027

Validation & Founding

- ▶ Peer-reviewed publication
- ▶ ESA BIC Incubation application
- ▶ Team building
- ▶ Seed round: \$2M–\$5M

PHASE 2

2028–2029

Satellite PDR & Partnerships

- ▶ Preliminary design review
- ▶ ESA Vigil data partnership
- ▶ Customer needs evaluation
- ▶ Series A: \$20M–\$50M

PHASE 3

2030+

Satellite CDR & Build

- ▶ Critical design review
- ▶ Flight qualification
- ▶ First launch target ~2033
- ▶ Essential utility status

DOCUMENTATION & TECHNICAL STACK



Technical Whitepaper

Complete system architecture, validation and analysis

- ▶ Appendix A: Orbital Mechanics (CR3BP dynamics)
- ▶ Appendix B: AI/ML Pipeline (architecture, training, validation)



GitHub Repository

Open-source · Fully reproducible · Documented

- ▶ `helios_code/` — Detection, triangulation, ensemble propagation
- ▶ `NeuralNetwork_ML/` — Model architecture, training, TMR voting
- ▶ `helios_orbits/` — CR3BP dynamics, differential correction, SRP
- ▶ `output/` — Trained model + validated results (JSON)

Technology Stack

AI/ML

Python · PyTorch
NumPy · SciPy
LLM verification

Physics Simulation

MATLAB/GMAT
CR3BP Dynamics
Monte Carlo

Orbital Mechanics

Differential Correction
Monodromy Matrix
Tsiolkovsky Eq.

Validation

6 Historical CMEs
3-Seed Ensemble
No Data Leakage

MVP Validation Pipeline

Input: Historical CME data → Feature engineering (16D vector) → Ensemble inference (3 seeds) → TMR consensus → Dose calculation → Severity classification → Output: Warning JSON

`run_complete_mvp.py`: Gold-standard script producing all whitepaper metrics — independently reproducible by any reviewer

TEAM & MILESTONES

FOUNDER

Paweł Micał

17-year old student

- ▶ Sole architect of the core system
- ▶ Conceptualized the physics-informed ML pipeline and fault-tolerant system design
- ▶ Validated MVP: 18-page whitepaper + 2 appendices + open-source code
- ▶ Expert feedback from Dr. Martin Lo (Genesis mission) on Halo Orbit Insertion values

MILESTONES

- ✓ **MVP Validation**
83.3% severity accuracy, 5.2 nT Bz error
- ✓ **Technical Whitepaper**
18 pages + 2 appendices
- ✓ **Open-Source Repository**
Full validation pipeline on GitHub
- ✓ **Orbital Analysis**
L1 halo orbit, ΔV budget optimized
- ✓ **Expert Consultation**
Dr. Martin Lo (Genesis) — Predicted HOI confirmed
- **Q2 2026: Publication**
Peer-reviewed journal submission
- **Q3 2026: ESA BIC Incubation**
Phase I application (€100K)
- **Q2 2027: Seed Round**
\$2M–\$5M for team + satellite PDR
- **Q1 2028: ESA Partnership**
ESA Vigil: Early Data Pipeline Integration

THE VISION

"By 2035, every human beyond Earth's magnetosphere will have HELIOS protection."

THE ASK

Seed Funding: \$2M – \$5M

- 40%** Engineering — satellite preliminary design, ground segment
- 30%** Team — CTO, BD lead, 2-3 engineers
- 20%** Business Development — customer pilots, conferences
- 10%** Operations — legal, IP protection, regulatory

KEY NUMBERS

- \$2.5B** Market by 2035
- 83.3%** Severity accuracy (validated)
- 16-50h** Warning lead time
- 5.2 nT** Bz prediction accuracy
- \$30M** Projected Year 10 revenue

"HELIOS isn't just a satellite constellation. It's the difference between life and death for the first Mars crew."